

Integrals – 6

Indefinite integrals: algorithms and non-elementarity

1 сентября 2026 года

Theoretical minimum.

Theorem 1 (Ostrogradsky's method). For a proper rational fraction $\frac{P}{Q}$ with $Q = Q_1 Q_2$, where $Q_1 = \gcd(Q, Q')$,

$$\int \frac{P}{Q} dx = \frac{P_1}{Q_1} + \int \frac{P_2}{Q_2} dx,$$

where P_1, P_2 are found from a linear system without computing the roots of Q ; the remaining integral has only simple poles.

Theorem 2 (Chebyshev on binomial differentials). $\int x^m(a+bx^n)^p dx$ is elementary if and only if one of the numbers $p, \frac{m+1}{n}, \frac{m+1}{n} + p$ is an integer; in these three cases the substitutions $x = t^N$, $a + bx^n = t^s$ and $ax^{-n} + b = t^s$ respectively do the job.

Theorem 3 (Liouville). Let f and g be rational functions with g non-constant. Then $\int f e^g dx$ is elementary if and only if there is a rational R with $f = R' + Rg'$. In particular $\int e^{x^2} dx$, $\int \frac{e^x}{x} dx$, $\int \frac{\sin x}{x} dx$ and $\int \frac{dx}{\ln x}$ are not elementary.

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1. Explain Ostrogradsky's method and apply it to $\int \frac{dx}{(x^2+1)^3}$ and to $\int \frac{x^4}{3-6x+6x^2-4x^3+2x^4} dx$ (Integration Bee 2024).
 2. Prove the sufficiency part of Chebyshev's theorem (the three substitutions) and decide which of the following are elementary: $\int x^3(1+x^2)^{1/2} dx$, $\int \frac{dx}{x^4\sqrt{1+x^2}}$, $\int \sqrt[3]{1+x^4} dx$, $\int x^{1/2}(1+x^{1/3})^{1/4} dx$.
 3. Reduce $\int R(x, \sqrt{ax^2+bx+c}) dx$ to the three canonical cases $\int R(t, \sqrt{1-t^2})$, $\int R(t, \sqrt{t^2-1})$, $\int R(t, \sqrt{t^2+1})$, and compute $\int \frac{x dx}{\sqrt{x^2-2}}$ (Integration Bee 2013) and $\int \frac{dx}{\sqrt{x(1-x)}}$ (Integration Bee 2017).

4. Prove that $\int e^{x^2} dx$ is not elementary following Liouville's scheme, and explain why the same argument proves the non-elementarity of $\int \frac{e^x}{x} dx$ but fails for $\int \frac{e^x}{x^2} dx$ – express the latter through the former.
5. (Integration Bee) A derivative inside out: recognise a total derivative in the integrand and compute
 - a) $\int ((2e^{x^2}x + 1) \cos x - (e^{x^2} + x) \sin x) dx$ (2016);
 - b) $\int e^x \left(\frac{1}{x} + \log x \right) dx$ (2018);
 - c) $\int (2 \log x + 1) e^{(\log x)^2} dx$ (2023);
 - d) $\int (x + e + 1) x^e e^x dx$ (2023);
 - e) $\int \cos^x(x) (\log(\cos x) - x \tan x) dx$ (2024).